

Technical Disclosure Commons

Defensive Publications Series

22 Mar 2023

Screenshots Enriched with Metadata to Enable Bookmarking and Semantic Retrieval

Duc-Hieu Tran

Follow this and additional works at: https://www.tdcommons.org/dpubs_series

Recommended Citation

Tran, Duc-Hieu, "Screenshots Enriched with Metadata to Enable Bookmarking and Semantic Retrieval", Technical Disclosure Commons, (March 22, 2023)
https://www.tdcommons.org/dpubs_series/5750



This work is licensed under a [Creative Commons Attribution 4.0 License](https://creativecommons.org/licenses/by/4.0/).

This Article is brought to you for free and open access by Technical Disclosure Commons. It has been accepted for inclusion in Defensive Publications Series by an authorized administrator of Technical Disclosure Commons.

Screenshots Enriched with Metadata to Enable Bookmarking and Semantic Retrieval

ABSTRACT

Users capture screenshots of their devices while running particular applications for various purposes such as bookmarking a specific interaction or state of the application. However, currently, finding the correct screenshot at a later time is challenging since screenshots are not amenable to searching. Further, the non-interactive nature of screenshots makes them inadequate for accessing underlying content such as embedded links or multimedia resources. This disclosure describes techniques to automatically enhance screenshots by capturing relevant contextual metadata that can enable semantic indexing of screenshots. The indexing facilitates search and retrieval and can support resuming the underlying application in the same state as at the time of screenshot capture. Semantic indexing can be performed using natural language understanding, image processing, optical character recognition, etc. The semantic index is usable to retrieve screenshots that match a user query and to automatically invoke the relevant application and resume it from the state captured in the screenshot.

KEYWORDS

- Enriched screenshot
- Enhanced screenshot
- Screenshot indexing
- Semantic indexing
- Application state
- Application view hierarchy
- Application package ID

BACKGROUND

Users often take screenshots of the content on their device screen. The screenshots can serve a variety of purposes, such as sharing the screen contents with a remote party, saving the information on the screen for later reference, etc. Since screenshots are captured as image files, the content within the image is not easily amenable to searching. As a result, users find it challenging to find specific screenshots at a later time. Even when users find the relevant screenshots, these are often difficult to use because content in the image is not interactive unlike the original content that was captured in the screenshot.

In some cases, users employ screenshots as a mechanism for bookmarking a specific interaction or state of an application, similar to saving webpage uniform resource locators (URLs) as bookmarks in web browsers. While users can pause some applications in a given state and/or recall content from specific history of their interaction within such applications, this functionality is typically rather limited compared to explicit bookmarking and is not available in all applications. For instance, in many messaging applications, users cannot bookmark specific sections within long chat sessions. Moreover, unlike web browser bookmarks, users cannot easily share the paused application state and/or interaction history with other parties.

Although taking screenshots as a workaround can overcome some of these limitations, users cannot return to the captured state of the application and continue their work simply from the screenshot, enabling the screenshot to act as a substitute for a bookmark. While a screenshot can typically be shared with others, the image format is often inadequate for sharing the underlying content. For instance, the viewer of a screenshot of an interaction within a messaging application cannot click on links or watch videos embedded in the screenshot of the interaction.

DESCRIPTION

This disclosure describes techniques, implemented with user permission, to enhance the screenshot images taken from applications on a device by capturing relevant metadata when users capture the screenshots. The metadata can enable semantic indexing to facilitate later search and retrieval of the desired screenshot from a collection of screenshots. The metadata associated with a given screenshot image can also be used to reconstruct the state of the underlying application at the time at which the user captured the screenshot. As a result, the techniques make it possible for users to bookmark a specific application state with a screenshot and easily revisit the same content later, e.g., by retrieving the screenshot with a search query.

With user permission, metadata regarding the current device and application context is captured and saved at the time of capture of a screenshot. The user can turn off capture of metadata for individual screenshots and/or applications, and also limit the metadata fields that are saved. As applicable, such metadata can include a variety of relevant information, such as:

- name or package ID of the application(s)
- URL(s) of webpage(s)
- on-screen text content
- related off-screen text content (e.g., parts of a current page that are not displayed on the screen because of limitations of screen size)
- the user interface (UI) elements of the applications visible on screen of those extracted from the view hierarchy along with their current states
- the current state of the application(s), such as filled in values of form fields, options in dialog boxes, text in notifications, etc.
- information regarding the context, such as date, time, location, etc.

The screenshot and associated metadata can be semantically indexed. Semantic indexing can be performed by processing text metadata with suitably trained natural language processing (NLP) and/or natural language understanding (NLU) models. The screenshot images and image content within the captured metadata can be represented via a dense vector representation that is processed via suitable image processing models, including extraction of text content within the images via optical character recognition (OCR). Application of OCR ensures that text content within the images that is not presented in the captured metadata is taken into account in the semantic indexing. The semantic indexing of screenshots and associated metadata yields a combined sparse index.

The semantic index generated as above can be used to retrieve results when the user issues a query to search for a specific screenshot at a later time. The search results can include the screenshot along with the associated metadata. The user can automatically invoke the relevant application(s) via the screenshot and/or the associated metadata. For instance, retrieving a previously captured screenshot of a playlist of songs playing within a music application can result in executing the music application based on recognizing the application ID within the metadata and/or the application name within the UI elements captured in the screenshot image.

Further, the user can choose to resume the application(s) in the state(s) at the time of taking the screenshot. For example, the image and/or the associated metadata of the screenshot of a music playlist can be used to invoke the music application with the same list of songs and start playback of the song playing at the time of screenshot capture at the same location within the song. Similarly, resuming a web browser application from a screenshot of specific parts of an article can result in opening the URL of the article in the web browser application and scrolling to the part of the article captured in the screenshot.

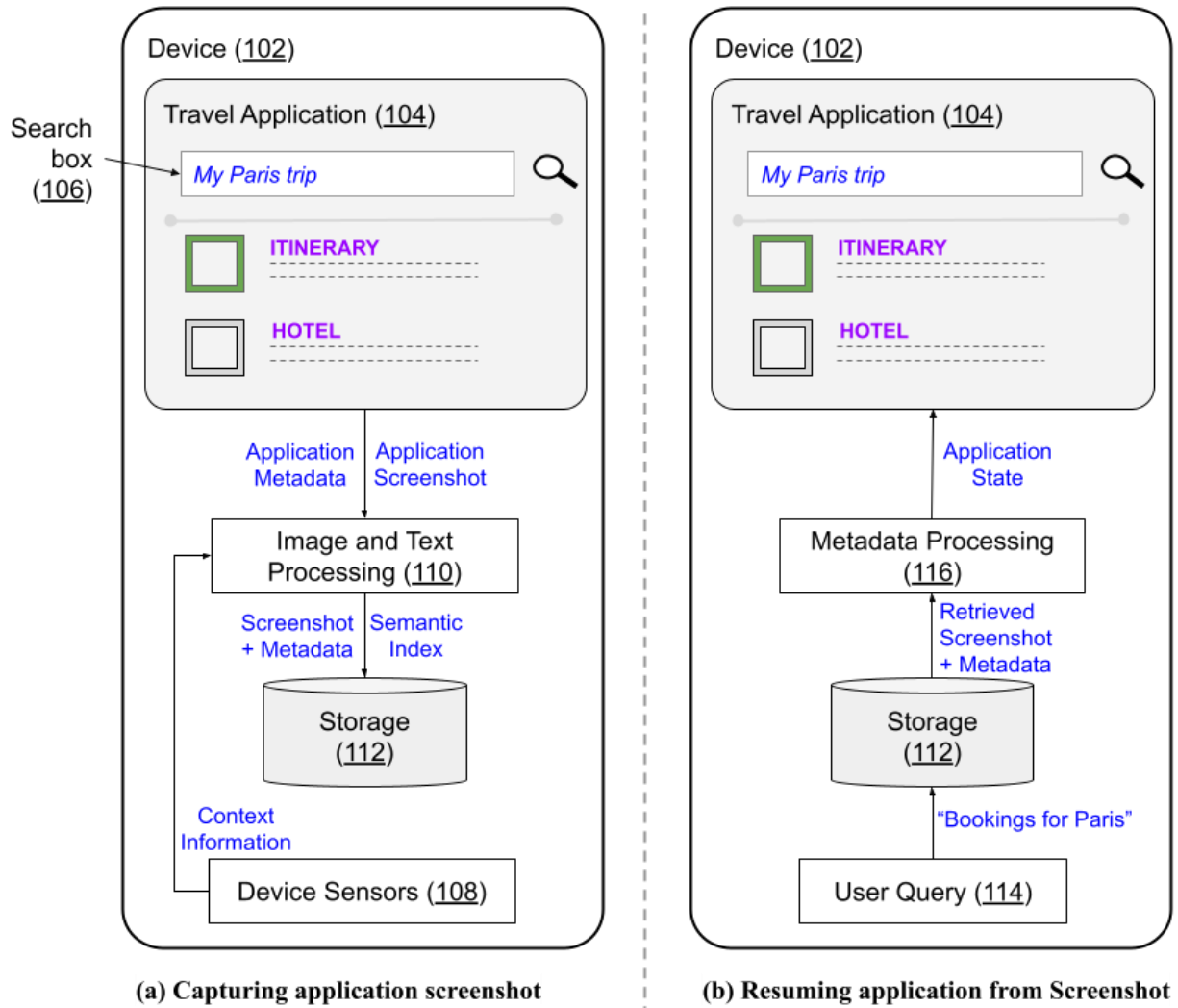


Fig. 1: Resuming an application in a previous state from captured screenshot and metadata

Fig. 1 illustrates an example of operational implementation of the techniques described in this disclosure. As shown in Fig. 1(a), a user captures a screenshot of a travel application (104) on a device (102) showing the results of the query entered in the search box (106) within the application. With user permission, metadata about the application and context information obtained via the device sensors (108) is captured along with the screenshot and is passed for processing via suitably trained models (110) for analyzing the image and text content within the

screenshot and the metadata. The combined output of the model yields a semantic index that is saved in storage (112) along with the screenshot and associated metadata.

As shown in Fig. 1(b), at a later time, the user issues a query (114) (“Bookings for Paris”) to search for the specific screenshot in the storage. The semantic index is used to retrieve the matching screenshot and associated metadata. The metadata is processed (116) to obtain relevant information about the application state when the user captured the screenshot. The application state is then passed to the application to enable resuming the application in the same state as at the time of screenshot capture.

In cases where applications cannot be resumed precisely in the specific state captured in screenshots, the application state can be restored as close as possible to the state within the screenshot. For instance, a screenshot of a search results page can be resumed with the same search query but latest results that may differ from those shown when the user captured the screenshot. The ability to take screenshots to bookmark a specific state within an application can be provided with any suitable user interface mechanism, such as a dedicated bookmarking button or menu option. Alternatively, or in addition, bookmarking functionality can be activated automatically whenever users take screenshots within an application. The captured device and application metadata can be embedded within the metadata of the associated screenshot. The metadata embedded within the screenshot can be appropriately encrypted to be accessible only to authorized users. For example, the metadata embedded in the screenshot of a chat application can be encrypted such that only users who are part of the chat can decrypt and access it.

The described techniques can be incorporated in a device operating system, a web browser, a messaging application, or other applications. The techniques can be employed to enhance screenshots by capturing relevant metadata on any device and within any application.

Semantic processing of screenshots and associated metadata can be performed with models running locally on the device. Alternatively, or in addition, and with appropriate user permissions, a part of the processing can be carried out externally, such as on a cloud platform or service. Similarly, the screenshots, associated metadata, and the generated semantic index can be stored locally on the device and/or externally on a server, in accordance with user settings. Implementation of the techniques can make it easier and faster to search, retrieve, and share screenshots and associated application states, thus enhancing the utility of screenshots and improving the user experience (UX) of saving, retrieving, and resuming application states via screenshots.

Further to the descriptions above, a user may be provided with controls allowing the user to make an election as to both if and when systems, programs or features described herein may enable collection of user information (e.g., information about a user's applications, screenshots, a user's context, social network, social actions or activities, profession, a user's preferences, or a user's current location), and if the user is sent content or communications from a server. In addition, certain data may be treated in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be treated so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a city, ZIP code, or state level), so that a particular location of a user cannot be determined. Thus, the user may have control over what information is collected about the user, how that information is used, and what information is provided to the user.

CONCLUSION

This disclosure describes techniques to automatically enhance screenshots by capturing relevant contextual metadata that can enable semantic indexing of screenshots. The indexing facilitates search and retrieval and can support resuming the underlying application in the same state as at the time of screenshot capture. Semantic indexing can be performed using natural language understanding, image processing, optical character recognition, etc. The semantic index is usable to retrieve screenshots that match a user query and to automatically invoke the relevant application and resume it from the state captured in the screenshot.

REFERENCES

1. Walnycky, Daniel, Ibrahim Baggili, Andrew Marrington, Jason Moore, and Frank Breitingner. "Network and device forensic analysis of android social-messaging applications." *Digital Investigation* 14 (2015): S77-S84.
2. Apple Inc. "App Metadata Specification 6.0." available online at <https://help.apple.com/asc/appsspec/en.lproj/static.html>, accessed Feb 20, 2023.
3. Chiatti, Agnese, Mu Jung Cho, Anupriya Gagneja, Xiao Yang, Miriam Brinberg, Katie Roehrick, Sagnik Ray Choudhury, Nilam Ram, Byron Reeves, and C. Lee Giles. "Text extraction and retrieval from smartphone screenshots: Building a repository for life in media." In *Proceedings of the 33rd Annual ACM Symposium on Applied Computing*, pp. 948-955. 2018.
4. Budurean, Cristina Elena, Richard Michael Gaywood, Dino Hughes, Johannes Tonollo, Philip James Adams, and Fergus Gerard Hurley. "Determining application test results using screenshot metadata." U.S. Patent 10,387,292, issued August 20, 2019.

5. Hu, Donghan, and Sang Won Lee. “Scrapbook: Screenshot-Based Bookmarks for Effective Digital Resource Curation across Applications.” In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, pp. 1-13. 2022.